



MINING DEALER BEST PRACTICE

Improving Ripper Cylinder Performance thru D11

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

Improving Ripper Cylinder Performance thru D11	1
1.0 Introduction.....	2
2.0 Best Practice Description	2
3.0 Implementation Steps.....	4
4.0 Benefits	4
5.0 Resources Required.....	5
6.0 Supporting Attachments / References	5
7.0 Related Best Practices	5
8.0 Acknowledgements	6

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1.0 Introduction

In iron ore mines, Ground Engaging Tools (GET) represent significant owning costs since wear rates are very severe on these operations. As per Caterpillar Handbook of Ripping (media number AEDK0752), a very critical factor on ripper tip wear rate and ripping production are track type tractor ripping operating techniques, specifically referring to shank position during ripping pass.

Literature and best operational practices recommend that the frame should be parallel to the ground, tracks flat on the ground, and shank slightly forward of vertical. In iron ore mines, operational efficiency teams have adopted as convention that all these conditions will be met if ripper tilt cylinder rods are 60% extended.

Even though these techniques are widely used, it is hard for the operator to visually define this optimum cylinder rod extension. The objective of this best practice is to give the operator an easy-to-see reference of this recommended cylinder extension using a rod attached to ripper tilt cylinder heads.

2.0 Best Practice Description

This best practice consists of the production of rod type tools, with length equal to 60% of rod extension, design showed in Figure 1, to be fastened to ripper tilt cylinder heads, using its own bolts, as per Figure 2. Rod length must be changed according to machine model/prefix and/or cylinder PN. Table 1 describes some of the possible lengths for the most common track type tractor models on mining applications.

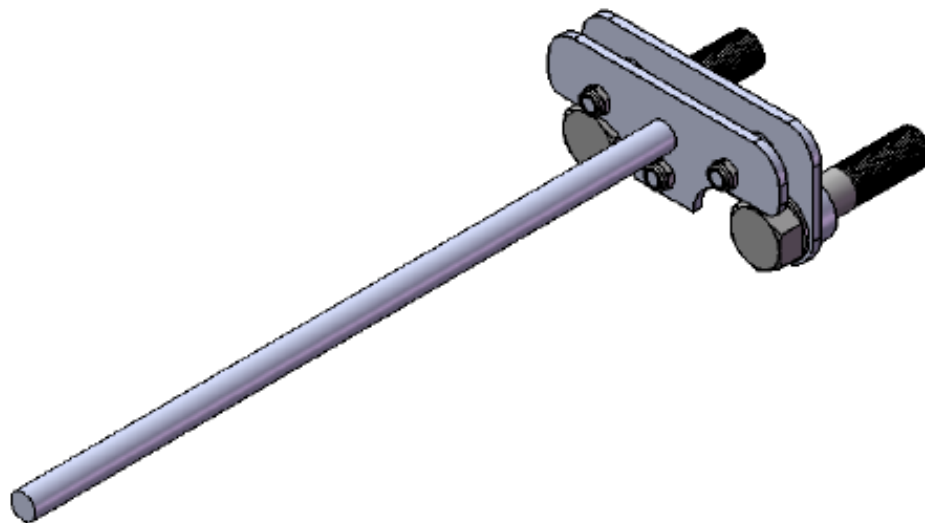


Figure 1 - Rod design

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Figure 2 - Rod fastened

Model	Prefix	Ripper Tilt Cyl. PN*	Fully extended rod length (mm)	60% extended rod length (mm)
D11	KSN	465-1711	830.00	498.00
D11T	AMA	130-3263 (RH) 130-6381 (LH)	830.00	498.00
D10	76X	3G-2220	818.60	491.16
D10T2	RAB	374-2285	660.00	396.00
D9	J3D	521-8391	456.90	274.14
D9T	TWG	521-8391	456.90	274.14

*Only considering single shank arrangements.

After installation, operators will have a view of the ideal cylinder rod extension, as shown in Figure 3.

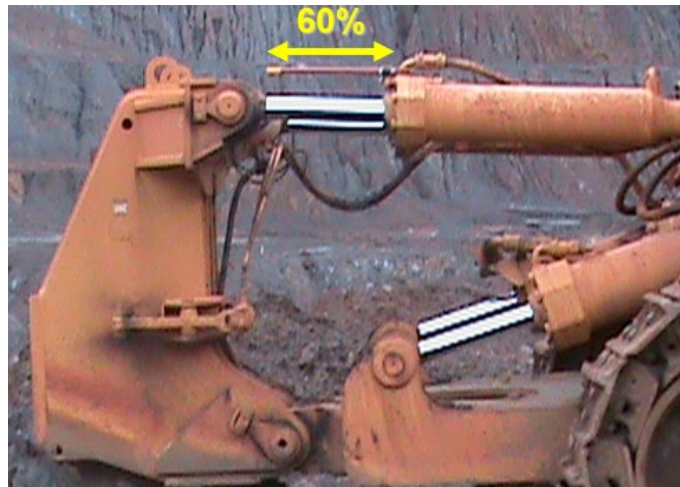


Figure 3 - Shank correctly positioned according to rod reference

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3.0 Implementation Steps

1. Develop specific designs for each target machine on the project
2. Manufacture rod tools for each target machine
3. Install the tools
4. Train operators
5. Field following

4.0 Benefits

In the current conditions, basing the cutting angle only on the operator's perception, ripper tips wear in a dome-shaped pattern, as shown in Figure 4.



Figure 4 - Dome shaped wear pattern

Using the correct techniques will allow for more uniform and sharp wear of ripper tips, lowering GET cost per hour. The difference is clearly visible when comparing Figure 4 with Figure 5, which shows the desirable wear pattern.

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Figure 5 - Sharp shaped wear pattern

Using this reference for operating techniques improvement led to an increase of around 20% of ripper tips lifetime. Additionally, to the parts cost savings associated, it represented higher physical availability and lower medium time between stops of track type tractors fleet on the tested site. Correct ripping removes the load from the rear idler and reduces track spin, improving undercarriage life.

5.0 Resources Required

- Third party partner to manufacture the tools
- Labor hours to install the tools

6.0 Supporting Attachments / References

- AEDK0752
- PEXT9050

7.0 Related Best Practices

N.A.

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Improving Ripper Cylinder Performance thru D11	DATE 7/29/2022	CHG NO 00	NUMBER 2.06-220729-1

8.0 Acknowledgements

This Best Practice was written by:

Jefferson Netto

SOTREQ S/A

E-mail: jefferson.netto@sotreq.com.br

Mobile: +55 031 9 9967-7110

Camila Rendeiro

SOTREQ S/A

E-mail: camila.rendeiro@sotreq.com.br

Mobile: +55 091 9 8146 9575

Thais Garcia

SOTREQ S/A

E-mail: thais.garcia@sotreq.com.br

Mobile: +55 94 8131-2676

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